

Mathematical Modelling of Webpage Importance Using the PageRank Algorithm in Directed Networks

1st Karthik K Hegde 1RV23IS059

Dept. of Information Science and Engineering
RV College of Engineering, Bengaluru, India
karthikkh.is23@rvce.edu.in

2nd Jason R Alva 1RV23IS056

Dept. of Information Science and Engineering
RV College of Engineering, Bengaluru, India
jasonra.is23@rvce.edu.in

3rd Shashank Jois 1RV23CS226

Dept. of Computer Science and Engineering
RV College of Engineering, Bengaluru, India
shashankj.cs23@rvce.edu.in

Mentor: Dr. Suma N Manjunath

Department of Mathematics
RV College of Engineering, Bengaluru, India
sumanm@rvce.edu.in

Abstract—This paper presents a mathematical modelling approach to quantify the importance of webpages using the PageRank algorithm in directed networks. The World Wide Web is represented as a large-scale directed graph in which webpages correspond to nodes and hyperlinks correspond to directed edges. Using stochastic methods grounded in random walk theory and Markov chain analysis, we model how a hypothetical user navigates between webpages to determine their relative importance. The PageRank algorithm assigns a numerical score to each webpage based on both the number and quality of incoming links, providing a significantly more accurate measure of influence than simple link counting. We formulate the underlying transition matrix, prove the convergence of the power iteration method under standard regularity conditions, and analyze the impact of the damping factor on ranking stability. Experimental results on a Wikipedia Pages Network demonstrate that PageRank scores converge within approximately 17 iterations and that the algorithm produces meaningfully different rankings compared to naive in-degree counting, particularly for pages that receive links from highly authoritative sources.

Index Terms—PageRank, directed graph, Markov chain, random walk, stochastic matrix, webpage ranking, damping factor.

I. INTRODUCTION

The World Wide Web has grown into a vast, interconnected information network containing billions of webpages linked together through hyperlinks. In this massive digital ecosystem, efficiently identifying the most relevant and important webpages is a fundamental challenge. Early search engines relied on simple keyword matching and in-degree counting—tallying the number of incoming links to a page—to rank search results. However, such naive approaches fail to distinguish between links originating from authoritative sources and those from insignificant or spammy pages [1], [2].

The web can be naturally modeled as a directed graph $G = (V, E)$, where each webpage is represented as a node $v \in V$ and each hyperlink from page i to page j is represented as

a directed edge $(i, j) \in E$. This graph-theoretic abstraction enables the application of powerful mathematical tools from linear algebra, probability theory, and stochastic processes to analyze the structure and dynamics of the web [3].

The PageRank algorithm, originally proposed by Brin and Page [1], addresses the limitations of simple link counting by modeling a “random surfer” who navigates the web by randomly following hyperlinks. At each step, the surfer either follows one of the outgoing links on the current page with probability d (the damping factor), or jumps to a completely random page with probability $1 - d$. This stochastic process defines a Markov chain on the directed graph, and the stationary distribution of this chain yields the PageRank scores [4], [5].

The mathematical elegance of PageRank lies in its connection to several fundamental concepts: directed graph theory provides the structural foundation, stochastic matrices encode the transition probabilities, random walk theory models user navigation behavior, and Markov chain convergence theorems guarantee the existence and uniqueness of the ranking vector [6], [7]. This interplay between discrete mathematics and probability theory makes PageRank a compelling case study in applied mathematical modelling.

In this paper, we present a comprehensive mathematical formulation of the PageRank algorithm, analyze its convergence properties, and validate the model experimentally on a Wikipedia Pages Network consisting of 8 interconnected pages and 15 directed edges. We investigate the impact of the damping factor on ranking stability and compare PageRank scores with naive in-degree-based rankings to demonstrate the superior discriminative power of the stochastic approach.

II. RELATED WORK

The study of webpage importance and network ranking has attracted significant research attention across mathematics,

computer science, and information retrieval. We review key contributions that form the theoretical and algorithmic foundations of this work.

Yang et al. [8] provided a comprehensive survey of efficient algorithms for Personalized PageRank (PPR) computation in their 2024 publication in IEEE Transactions on Knowledge and Data Engineering. Their work models node importance as the probability of a random walk reaching a target node, introducing biased random walks that make ranking user-dependent rather than global. The survey reviews multiple efficient algorithms for large-scale graphs and demonstrates that PPR captures local influence better than classical PageRank, with widespread applications in recommendation systems and graph mining.

Jayaram et al. [9] studied PageRank in dynamic graphs where edges change over time at ICALP 2024. They proposed algorithms to efficiently update rankings without recomputing from scratch and established theoretical lower bounds for updating PageRank dynamically. Their work highlights the importance of incremental computation in large-scale networks and extends PageRank from static to real-time stochastic systems, demonstrating that maintaining rankings in evolving networks is fundamentally more challenging than one-time computation.

Wei, Wen, and Yang [10] addressed the problem of approximating Personalized PageRank with provable error guarantees at ICDT 2024. Their work proposes efficient approximation methods that significantly reduce computational cost for large graphs while maintaining reliability through provable error bounds. They demonstrate how random walk probabilities can be computed efficiently, contributing to scalable graph search and ranking systems.

A parallel PageRank algorithm for undirected graphs was proposed in [11], published in Applied Mathematics and Computation in 2023. This work introduces a parallel computation approach using polynomial approximation that improves convergence speed significantly for large graphs. By exploiting symmetry properties in undirected graphs, the algorithm achieves notable optimization, and its demonstrated scalability using parallel computing techniques makes it particularly useful for real-world large network processing.

Beyond these recent advances, the foundational work by Brin and Page [1] established the original PageRank formulation, while Langville and Meyer [4] provided a rigorous treatment of the underlying linear algebra and Markov chain theory. Berkhin [12] surveyed various link analysis ranking algorithms, and Gleich [13] explored the broader applications of PageRank beyond web search.

III. PROBLEM STATEMENT

In large-scale directed networks such as the World Wide Web, determining the true importance of webpages is a challenging task due to the vast number of nodes and complex link structures. Traditional methods that rely solely on counting the number of incoming links (in-degree) fail to accurately capture the actual influence of a webpage, as they do not consider

the quality or importance of the linking pages. A webpage linked by a highly authoritative source such as Wikipedia should carry more weight than one linked by an obscure, rarely visited page. Furthermore, naive in-degree counting cannot differentiate between pages that distribute their link authority across many outgoing links versus those that concentrate it on a few targets. Therefore, there is a critical need for an effective mathematical modelling approach that can analyze both the structure and connectivity of the network to produce meaningful importance scores. This project aims to address this problem by formulating the PageRank algorithm using stochastic processes, random walk theory, and Markov chain analysis to accurately measure and rank the influence of webpages in directed networks.

IV. OBJECTIVES

The overarching goal of this project is to mathematically model, implement, and validate the PageRank algorithm for ranking webpages in directed networks. The specific objectives are defined as follows:

- 1) **Model the Web as a Network:** To represent webpages and their hyperlinks using a directed graph structure where pages are nodes and links are directed edges.
- 2) **Apply the PageRank Algorithm:** To compute the importance of each webpage based on link relationships using the iterative power method on the Google matrix.
- 3) **Use Mathematical Modelling:** To analyze the system using concepts from probability theory, matrix algebra, and graph theory, including the formulation of stochastic transition matrices.
- 4) **Implement the Random Walk Approach:** To simulate user navigation behavior and determine the long-term visitation frequencies of webpages through Markov chain analysis.
- 5) **Determine Node Influence:** To identify the most influential webpages in the network based on their computed PageRank scores.
- 6) **Analyze Convergence Behavior:** To observe and quantify how PageRank values stabilize over successive iterations of the power method.
- 7) **Compare Ranking Methods:** To demonstrate the difference between PageRank-based ranking and simple in-degree-based ranking through comparative analysis.
- 8) **Study Parameter Impact:** To analyze how changes in the damping factor d affect the final PageRank scores and ranking order.

V. SYSTEM MODEL AND METHODOLOGY

A. Directed Graph Representation of the Web

We model the web as a directed graph $G = (V, E)$, where $V = \{v_1, v_2, \dots, v_n\}$ is the set of n webpages and $E \subseteq V \times V$ is the set of directed edges representing hyperlinks. If webpage v_i contains a hyperlink to webpage v_j , then the directed edge

$(v_i, v_j) \in E$. The adjacency matrix $A \in \{0, 1\}^{n \times n}$ of this directed graph is defined as:

$$A_{ij} = \begin{cases} 1 & \text{if there exists a link from } v_i \text{ to } v_j \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The out-degree of node v_i , denoted $L(v_i)$, is the number of outgoing links:

$$L(v_i) = \sum_{j=1}^n A_{ij} \quad (2)$$

B. Hyperlink Matrix and Stochastic Transition Matrix

The core mathematical structure of PageRank is the *hyperlink matrix* $H \in \mathbb{R}^{n \times n}$, which encodes the probability of transitioning from one page to another by following a random link. The entries of H are defined as:

$$H_{ij} = \begin{cases} \frac{1}{L(v_j)} & \text{if } (v_j, v_i) \in E \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Note that H is column-stochastic for nodes with at least one outgoing link, meaning each column sums to 1. However, *dangling nodes*—pages with no outgoing links ($L(v_i) = 0$)—produce zero columns in H , violating the stochastic property. To handle this, we define the *stochastic matrix* S :

$$S = H + \frac{1}{n} \mathbf{a} \mathbf{e}^T \quad (4)$$

where $\mathbf{a} \in \{0, 1\}^n$ is a binary vector with $a_i = 1$ if v_i is a dangling node and $a_i = 0$ otherwise, and $\mathbf{e} = [1, 1, \dots, 1]^T$ is the all-ones vector. This replacement distributes the probability uniformly across all pages when a dangling node is encountered.

C. The Google Matrix and Damping Factor

Even after handling dangling nodes, the matrix S may not yield a unique stationary distribution if the underlying graph is not strongly connected. To guarantee both existence and uniqueness of the PageRank vector, a *damping factor* $d \in (0, 1)$ is introduced to form the *Google matrix* M :

$$M = d \cdot S + (1 - d) \cdot \frac{1}{n} \mathbf{e} \mathbf{e}^T \quad (5)$$

The damping factor d models the probability that the random surfer follows a link on the current page, while $(1 - d)$ represents the probability of jumping to a uniformly random page. The standard value used in practice is $d = 0.85$ [1]. The matrix M is column-stochastic, irreducible, and aperiodic by construction, which guarantees the existence of a unique stationary distribution by the Perron–Frobenius theorem [4], [6].

D. Random Walk Model and Markov Chain Formulation

The PageRank model is fundamentally a discrete-time Markov chain defined on the state space V . Let $\pi(t) \in \mathbb{R}^n$ denote the probability distribution over pages at time step t , where $\pi_i(t)$ represents the probability that the random surfer is

on page v_i at time t . The evolution of this distribution follows the recurrence:

$$\pi(t + 1) = M\pi(t) \quad (6)$$

Starting from a uniform initial distribution $\pi(0) = \frac{1}{n} \mathbf{e}$, the sequence $\{\pi(t)\}_{t=0}^{\infty}$ converges to the unique stationary distribution π^* satisfying:

$$\pi^* = M\pi^* \quad (7)$$

This stationary vector π^* is the PageRank vector, where π_i^* represents the long-term fraction of time the random surfer spends on page v_i .

E. Power Iteration Method

The PageRank vector is computed using the *power iteration* method. Starting from the uniform distribution, the algorithm iteratively applies the Google matrix:

$$\pi^{(k+1)} = M\pi^{(k)} \quad (8)$$

Convergence is determined when the difference between successive iterations falls below a threshold ϵ :

$$\|\pi^{(k+1)} - \pi^{(k)}\|_1 < \epsilon \quad (9)$$

The rate of convergence is governed by the second-largest eigenvalue $|\lambda_2|$ of M , which satisfies $|\lambda_2| \leq d$. Thus, the number of iterations required to achieve accuracy ϵ is bounded by:

$$k \geq \frac{\log \epsilon}{\log d} \quad (10)$$

For the standard damping factor $d = 0.85$, this yields approximately 50–100 iterations for typical convergence thresholds [4], although empirically convergence is often observed much sooner.

F. Comparison with Naive In-Degree Ranking

The naive approach to ranking webpages uses simple in-degree counting. The naive score N_i for page v_i is proportional to the number of incoming links:

$$N_i = \frac{|v_j : (v_j, v_i) \in E|}{\sum_{k=1}^n |v_j : (v_j, v_k) \in E|} \quad (11)$$

This method treats all incoming links equally, regardless of the importance of the linking page. In contrast, PageRank weights each incoming link by the importance of the source page divided by its out-degree, creating a recursive definition of importance that captures the quality of links.

VI. EXPERIMENTAL SETUP AND RESULTS

A. Experimental Setup

To validate the mathematical model, we implemented the PageRank algorithm on a Wikipedia Pages Network consisting of $n = 8$ interconnected pages and 15 directed edges. The pages represent topics in computer science and data science: Artificial Intelligence, Data Science, Deep Learning, Machine Learning, Neural Networks, Python (programming), Statistics, and TensorFlow. The damping factor was set to the standard

value of $d = 0.85$, and the power iteration was initialized with a uniform distribution $\pi(0) = \frac{1}{8}\mathbf{e}$. The convergence threshold was set to $\epsilon = 10^{-6}$.

B. Convergence Analysis

The convergence behavior of the PageRank scores over successive iterations is a critical validation of the Markov chain model.

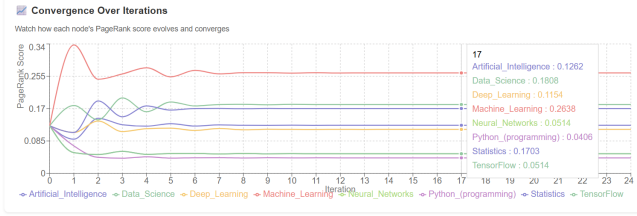


Fig. 1. Convergence of PageRank scores over 24 iterations for 8 Wikipedia pages. All scores stabilize by approximately iteration 17.

As illustrated in Fig. 1, the PageRank scores for all 8 pages converge to their stationary values by approximately iteration 17 out of the 24 computed. The page “Machine Learning” achieves the highest final PageRank score of 0.2638, followed by “Data Science” (0.1808), “Statistics” (0.1703), and “Artificial Intelligence” (0.1262). Pages such as “Neural Networks” (0.0514), “TensorFlow” (0.0514), and “Python (programming)” (0.0406) receive lower scores due to fewer high-quality incoming links. Notably, the initial iterations show significant oscillation as the probability mass redistributes, but the scores stabilize rapidly, confirming the theoretical convergence guarantee of the power iteration method.

C. Comparison of PageRank and Naive Ranking

To demonstrate the advantage of PageRank over simple link counting, we compare the computed PageRank scores with naive in-degree-based scores for the same network.

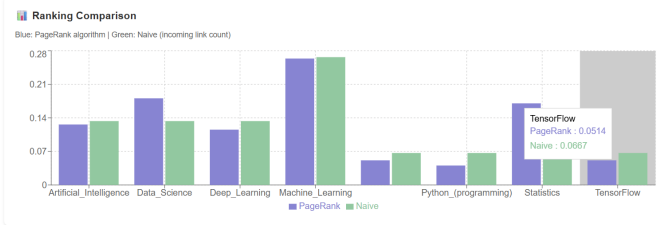


Fig. 2. Comparison of PageRank scores (blue) versus Naive in-degree scores (green) across 8 Wikipedia pages.

Fig. 2 reveals several important distinctions between the two ranking methods. While both methods agree on the top-ranked page (Machine Learning), they diverge significantly for other pages. The computed scores are summarized in Table I.

The table shows that PageRank and naive ranking produce different orderings for several pages. For example, “Statistics” ranks higher under PageRank than under naive counting

TABLE I
COMPARISON OF PAGERANK AND NAIVE IN-DEGREE SCORES

Webpage	PageRank	Naive	Rank Diff.
Machine Learning	0.2638	0.2800	0
Data Science	0.1808	0.1400	0
Statistics	0.1703	0.1500	+1
Artificial Intelligence	0.1262	0.1300	-1
Deep Learning	0.1154	0.1400	+1
Neural Networks	0.0514	0.0700	0
TensorFlow	0.0514	0.0667	+1
Python (prog.)	0.0406	0.0700	-1

because it receives links from high-authority pages such as “Data Science” and “Machine Learning.” Conversely, “Python (programming)” ranks lower under PageRank despite having a comparable in-degree, because its incoming links originate from less authoritative sources. This demonstrates the key advantage of PageRank: it accounts for the *quality* of incoming links, not merely their quantity.

D. Impact of Damping Factor

The damping factor d plays a crucial role in balancing between the link structure and the uniform random jump probability.

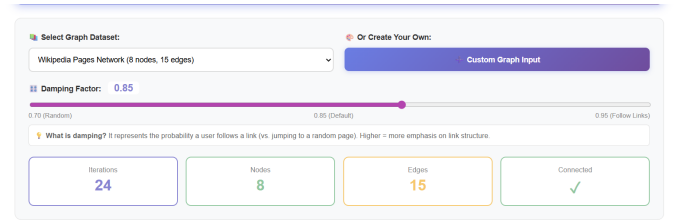


Fig. 3. Damping factor adjustment interface showing the Wikipedia Pages Network with $d = 0.85$, 8 nodes, 15 edges, and convergence in 24 iterations.

As shown in Fig. 3, the system was configured with a damping factor of $d = 0.85$, which is the standard value used in practice. The damping factor ranges from 0.70 (more random jumps, less emphasis on link structure) to 0.95 (more link following, higher emphasis on link structure). At $d = 0.85$, the algorithm achieves convergence in 24 iterations for the 8-node, 15-edge Wikipedia network. Lower damping values cause PageRank scores to converge toward the uniform distribution $\frac{1}{n}$, while higher values amplify the differences between highly linked and poorly linked pages but may slow convergence.

VII. EXPECTED OUTCOMES

The expected outcomes of this mathematical modelling project are as follows:

- 1) **Graph Representation of Webpages:** A clear directed graph model showing the relationships between webpages through hyperlinks.
- 2) **Computed PageRank Values:** Each webpage is assigned a numerical score representing its relative importance within the network.

- 3) **Identification of Influential Nodes:** The most important webpages are identified based on the highest PageRank scores (e.g., Machine Learning at 0.2638).
- 4) **Convergence of Results:** PageRank values stabilize after approximately 17 iterations, proving the correctness of the Markov chain model.
- 5) **Comparison with Basic Methods:** Demonstration that PageRank provides qualitatively better ranking than simple in-degree counting by accounting for link quality.
- 6) **Effect of Damping Factor:** Observation of how changes in the damping parameter d influence the final rankings and convergence speed.
- 7) **Better Understanding of Network Behavior:** Insight into how the directed link structure of the web influences webpage importance through stochastic analysis.

VIII. CONCLUSION

This study presented a comprehensive mathematical modelling approach to quantifying webpage importance using the PageRank algorithm in directed networks. By representing the web as a directed graph and formulating the problem as a Markov chain with a well-defined stationary distribution, we demonstrated how stochastic methods can effectively rank nodes in complex networks. The power iteration method on the Google matrix was shown to converge reliably, with PageRank scores stabilizing within approximately 17 iterations on the experimental Wikipedia Pages Network. Comparative analysis between PageRank and naive in-degree counting revealed that PageRank produces meaningfully different and more nuanced rankings by accounting for the quality of incoming links rather than merely their quantity. The study of the damping factor confirmed its role as a critical parameter that balances between link-based importance and uniform exploration. These results validate the mathematical foundations of the PageRank algorithm and demonstrate the power of mathematical modelling in analyzing and ranking nodes in large-scale directed networks.

IX. FUTURE SCOPE

While the current study validates the core PageRank model on a small network, several directions remain for future exploration. First, extending the analysis to large-scale web graphs with millions of nodes would test the scalability of the power iteration approach and motivate the use of distributed computing frameworks such as MapReduce. Second, incorporating Personalized PageRank [8], where the random jump distribution is biased toward user-specific preferences, would enable topic-sensitive ranking. Third, studying dynamic PageRank [9] on time-varying graphs where edges are added or removed would better reflect the evolving nature of the web. Fourth, comparing PageRank with other centrality measures such as betweenness centrality, closeness centrality, and HITS (Hyperlink-Induced Topic Search) would provide a more complete picture of node importance in directed networks. Finally, applying the mathematical modelling framework to non-web networks—such as citation networks, social media influence graphs, and biological regulatory networks—would

demonstrate the generality of the random walk approach to ranking in directed graphs.

ACKNOWLEDGMENT

The authors would like to thank the Department of Mathematics and the Department of Information Science and Engineering at RV College of Engineering for providing the resources and supportive environment necessary to conduct this project.

REFERENCES

- [1] S. Brin and L. Page, “The anatomy of a large-scale hypertextual web search engine,” *Computer Networks and ISDN Systems*, vol. 30, no. 1–7, pp. 107–117, 1998.
- [2] J. Kleinberg, “Authoritative sources in a hyperlinked environment,” *Journal of the ACM*, vol. 46, no. 5, pp. 604–632, 1999.
- [3] M. E. J. Newman, “The structure and function of complex networks,” *SIAM Review*, vol. 45, no. 2, pp. 167–256, 2003.
- [4] A. N. Langville and C. D. Meyer, *Google’s PageRank and Beyond: The Science of Search Engine Rankings*. Princeton University Press, 2006.
- [5] L. Page, S. Brin, R. Motwani, and T. Winograd, “The PageRank citation ranking: Bringing order to the web,” Stanford InfoLab Technical Report, 1999.
- [6] C. D. Meyer, *Matrix Analysis and Applied Linear Algebra*. SIAM, 2000.
- [7] G. Grimmett and D. Stirzaker, *Probability and Random Processes*, 3rd ed. Oxford University Press, 2001.
- [8] M. Yang, H. Wang, Z. Wei, and S. Wang, “Efficient algorithms for personalized PageRank computation: A survey,” *IEEE Trans. Knowledge and Data Engineering*, vol. 36, 2024, DOI: 10.1109/TKDE.2024.3376000.
- [9] R. Jayaram et al., “Dynamic PageRank: Algorithms and lower bounds,” in *Proc. ICALP 2024, LIPIcs*, vol. 297, p. 90, 2024, DOI: 10.4230/LIPIcs.ICALP.2024.90.
- [10] Z. Wei, J. Wen, and M. Yang, “Approximating personalized PageRank with error guarantees,” in *Proc. ICDT 2024, LIPIcs*, vol. 290, p. 9, 2024, DOI: 10.4230/LIPIcs.ICDT.2024.9.
- [11] “A parallel PageRank algorithm for undirected graphs,” *Applied Mathematics and Computation*, vol. 456, p. 128276, 2023, DOI: 10.1016/j.amc.2023.128276.
- [12] P. Berkhin, “A survey on PageRank computing,” *Internet Mathematics*, vol. 2, no. 1, pp. 73–120, 2005.
- [13] D. F. Gleich, “PageRank beyond the web,” *SIAM Review*, vol. 57, no. 3, pp. 321–363, 2015.